MIT Open Learning Library
Standard Course Extraction Template

Reusable template for all seven MIT courses feeding the QAI Enterprise Library

1. Course Identity

Course title: [Enter source-supported information]

MITx / course ID: [Enter source-supported information]

Term / course version: [Enter source-supported information]

MIT Open Learning Library URL: [Enter source-supported information]

Primary domain: [Enter source-supported information]

Course purpose: [Enter source-supported information]

Source material available: [Enter source-supported information]

Extraction status: [Enter source-supported information]

2. Course Fidelity and Source Structure

Preserve the actual MIT OLL organization. Do not replace MIT terminology with QAI terminology at this layer.

- Subunits / modules
- Lectures
- Problem sets
- Assessments / graded questions
- References
- Other course components

3. Course Map

Subunit	Item	Type	Questions	Source Available	Status

4. Module / Subunit Record

Subunit ID / name: [Enter source-supported information]

MIT source URL: [Enter source-supported information]

Learning objectives: [Enter source-supported information]

Lectures / topics: [Enter source-supported information]

Problem sets: [Enter source-supported information]

Questions: [Enter source-supported information]

Important concepts: [Enter source-supported information]

References: [Enter source-supported information]

5. Lecture / Content Record

Lecture / item ID: [Enter source-supported information]

Title: [Enter source-supported information]

Type: [Enter source-supported information]

MIT source URL: [Enter source-supported information]

Question count: [Enter source-supported information]

Concepts covered: [Enter source-supported information]

Key mathematical ideas: [Enter source-supported information]

Implementation relevance: [Enter source-supported information]

6. Problem Set Record

Problem set ID / title: [Enter source-supported information]

MIT source URL: [Enter source-supported information]

Question count: [Enter source-supported information]

Original problem available: [Enter source-supported information]

Solved questions available: [Enter source-supported information]

Notebook / code available: [Enter source-supported information]

Related lecture(s): [Enter source-supported information]

7. Solved Question Extraction

Use one record per question and preserve the original problem/solution relationship.

Question ID: [Enter source-supported information]

Problem statement: [Enter source-supported information]

Given information: [Enter source-supported information]

Required result: [Enter source-supported information]

Solution / reasoning: [Enter source-supported information]

Mathematical derivation: [Enter source-supported information]

Algorithm / procedure: [Enter source-supported information]

Key insight: [Enter source-supported information]

Assumptions: [Enter source-supported information]

Complexity: [Enter source-supported information]

Implementation clues: [Enter source-supported information]

Validation / result: [Enter source-supported information]

8. Notebook / Computational Asset Record

Notebook name: [Enter source-supported information]

Course / problem-set relationship: [Enter source-supported information]

Original source: [Enter source-supported information]

Language: [Enter source-supported information]

Framework / SDK: [Enter source-supported information]

Dependencies: [Enter source-supported information]

Input: [Enter source-supported information]

Output: [Enter source-supported information]

Algorithm / model: [Enter source-supported information]

Execution environment: [Enter source-supported information]

Result: [Enter source-supported information]

Validation: [Enter source-supported information]

Reusable pattern: [Enter source-supported information]

QAI relationship: [Enter source-supported information]

9. Algorithms and Mathematical Models

Record only what is supported by course, problem-set, solution, notebook, or experiment evidence.

Algorithm: [Enter source-supported information]

Mathematical model: [Enter source-supported information]

Inputs: [Enter source-supported information]

Outputs: [Enter source-supported information]

Complexity: [Enter source-supported information]

Implementation evidence: [Enter source-supported information]

Related problems: [Enter source-supported information]

Related notebooks: [Enter source-supported information]

10. Extracted Pattern

Pattern name: [Enter source-supported information]

Source course / module: [Enter source-supported information]

Source problems: [Enter source-supported information]

Problem-solving structure: [Enter source-supported information]

Inputs: [Enter source-supported information]

Outputs: [Enter source-supported information]

Control / feedback: [Enter source-supported information]

Recurring evidence: [Enter source-supported information]

Potential reuse: [Enter source-supported information]

Confidence / evidence level: [Enter source-supported information]

11. Primitive Candidate

Primitive name: [Enter source-supported information]

Source evidence: [Enter source-supported information]

Pattern: [Enter source-supported information]

Purpose: [Enter source-supported information]

Proposed inputs: [Enter source-supported information]

Proposed outputs: [Enter source-supported information]

API concept: [Enter source-supported information]

Runtime requirements: [Enter source-supported information]

Dependencies: [Enter source-supported information]

Validation required: [Enter source-supported information]

Maturity: [Enter source-supported information]

Promotion recommendation: [Enter source-supported information]

12. QAI Architecture Mapping

- QAI Language
- QAI Primitives
- QAI OS
- QAI Runtime
- Hybrid Runtime
- QAI Control Plane
- Quantum Control Plane
- Adaptive Network Fabric
- Resource Registry
- Capability Registry
- QAI Labs
- QAI Product Foundry

13. Classification

- Educational knowledge
- Research
- Mathematical model
- Algorithm
- Pattern
- Prototype
- Implementation evidence
- Reusable capability
- Future capability

14. Evidence and Provenance

MIT OLL source URL(s): [Enter source-supported information]

Original document / problem-set reference: [Enter source-supported information]

Notebook reference: [Enter source-supported information]

Solution reference: [Enter source-supported information]

External reference (if any): [Enter source-supported information]

Evidence limitations: [Enter source-supported information]

15. Cross-Course Convergence

Identify whether the same concept, algorithm, pattern, or primitive appears in other MIT courses.

Course	Topic / Problem	Common Pattern	Primitive Candidate	Evidence	Status

16. Reuse and Implementation Assessment

- Direct reuse
- Pattern reuse
- Design reference
- Research input
- Architecture input
- Prototype candidate
- New implementation required
- Historical / educational reference only

17. Validation Plan

Reference implementation needed: [Enter source-supported information]

Simulator validation: [Enter source-supported information]

Hardware / QPU validation: [Enter source-supported information]

Independent problem validation: [Enter source-supported information]

Cross-course validation: [Enter source-supported information]

Benchmark: [Enter source-supported information]

Acceptance criteria: [Enter source-supported information]

18. Final Course Summary

- Major concepts
- Important algorithms
- Important problem-solving patterns
- Important solved questions
- Important notebooks
- Primitive candidates
- Enterprise Library contributions
- QAI architecture contributions
- Implementation opportunities
- Evidence gaps

19. Extraction Status

Area	Status	Notes

Course structure	Pending	
Modules / subunits	Pending	
Lectures	Pending	
Problem sets	Pending	
Solved questions	Pending	
Notebooks	Pending	
Algorithms	Pending	
Mathematical models	Pending	
Patterns	Pending	
Primitive candidates	Pending	
QAI mapping	Pending	
Cross-course mapping	Pending	
Validation	Pending	
Final review	Pending	

20. Extraction Rules

- Preserve the actual MIT OLL hierarchy and terminology.
- Keep original problem statements and solved questions associated.
- Treat solved questions as first-class evidence.
- Treat notebooks as implementation evidence.
- Do not infer implementation maturity from educational material.
- Separate source evidence from QAI interpretation.
- Promote patterns to primitive candidates only when supported by evidence.
- Promote primitives toward the Enterprise Library only after appropriate validation.
- Use cross-course recurrence to strengthen, not replace, validation.